NWDC Toolbox

Standard inputs

Every download function uses these four inputs, in this order.

Input	Meaning	How to enter it
`location`	Where	State ("ID" or "Idaho"), county FIPS code ("16001"), or HUC code of any size ("17050114")
`start`	First date	Year (2015) or year-month ("2015-06")
`end`	Last date	Leave out to get only start period, leave out start date to get most recent period
`timestep`	Time grouping	"monthly" (default), "calendar_year" or "water_year" (Oct-Sep)

Results come back for every HUC12 watershed inside the location.

Available functions

`list_models(sector=None)`

Lists the datasets and the years they cover.

Input	What to enter
`sector`	Optional: "use", "quantity" or "availability"
Example	`nwdc.list_models()`

`list_variables(model=None, sector=None)`

Lists variable IDs, names, units and source models.

Input	What to enter
`model`	Optional: e.g. "prms"
`sector`	Optional: "use", "quantity" or "availability"
Example	`nwdc.list_variables(sector="use")`

`describe(name)`

Prints a short description of a variable or model.

Input	What to enter
`name`	Variable ID or model, e.g. "recharge"
Example	`nwdc.describe("recharge")`

get_water_use(sector, location, start, end, timestep, use, source)

Water use in million gallons per day.

Input	What to enter
sector	"irrigation", "public_supply" or "thermoelectric"
location, start, end, timestep	Standard inputs
use	"withdrawals" (default) or "consumptive_use"
source	"total" (default), "groundwater", "surface_water" or "all"
Example	nwdc.get_water_use("irrigation", "ID", 2015, 2020)

get_precipitation(location, start, end, timestep)

Precipitation in mm per month, 1979-2021.

Input	What to enter
location, start, end, timestep	Standard inputs
Example	nwdc.get_precipitation("ID", 2015)

get_hydrology(location, start, end, timestep, variables, model)

Modeled ET, runoff, baseflow, snow, soil moisture and recharge.

Input	What to enter
location, start, end, timestep	Standard inputs
variables	"all" (default) or a list: actet, incrunoff, incbsflow, incqkflow, swe, soilmstfr, soilmst, recharge
model	"ensemble" (default, 2009-2020), "prms" (1983-2021) or "wrf-hydro" (2009-2021)
Example	nwdc.get_hydrology("ID", 2015, model="prms")

get_water_availability(location, start, end, timestep, variables)

2025 assessment results: availability, streamflow, consumption, stress index. 2009-2020.

Input	What to enter
location, start, end, timestep	Standard inputs
variables	"all" (default) or a list: availab, strflow, consum, sui
Example	nwdc.get_water_availability("ID", 2015)

get_data(variables, location, start, end, timestep, model)

Downloads any variable by ID.

Input	What to enter
variables	One ID or a list from the same model, e.g. "pswdtot"
location, start, end, timestep	Standard inputs
model	Only needed for hydrology variables

Input	What to enter
Example	`nwdc.get_data("pswdtot", "ID", 2015)`